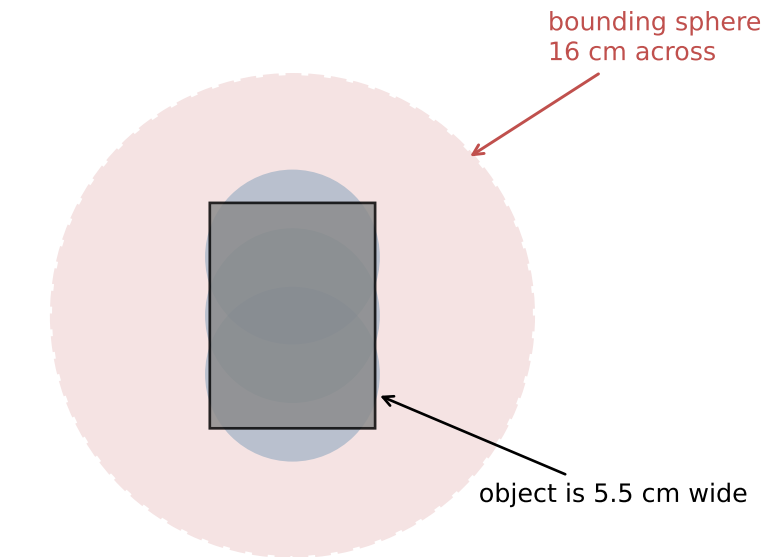
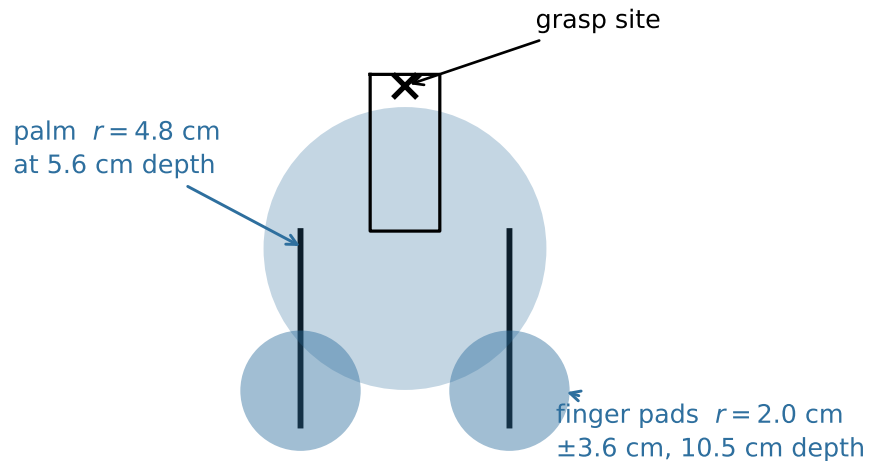


(a) Why the obstacle is decomposed



*decomposition (blue) tracks the shape;
the bounding sphere (red) inflates it threefold*

(b) The three-sphere end-effector model



Both panels are drawn to scale from the constants the shield uses. Arm links (not shown) carry one radius each, sampled at three points along the link axis: a coarser representation than the end-effector, which is where Section 4.5 locates the residual collisions.